

PADHAISHURU

GATE Preparation Platform

GATE MT

Metallurgical Engineering

TREND-BASED MOCK PAPER

— 03 —

Duration	3 Hours
Total Questions	65
Maximum Marks	100
Sections	10 subjects

Difficulty Distribution

Easy: 23 questions (35%)

Moderate: 33 questions (51%)

Difficult: 9 questions (14%)

Unofficial practice material. Not affiliated with or endorsed by GATE/IIT.
These mock papers are constructed using analysis of historical GATE
PYQ patterns.
Historical patterns do not guarantee future examination questions.

GENERAL INSTRUCTIONS

GATE MT — Trend-Based Mock Paper

1. Examination Pattern

The paper contains 65 questions carrying a total of 100 marks. The duration of the examination is 3 hours.

2. Question Types

Questions are of three types:

- (a) Multiple Choice Questions (MCQ): One correct answer out of four options.
 - (b) Multiple Select Questions (MSQ): One or more correct answers out of four options. Each correct answer carries full marks. There is no negative marking.
 - (c) Numerical Answer Type (NAT): A real number answer to be entered. No options are provided.
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3. Marking Scheme

- 1-mark questions: +1 for correct, -0.33 for incorrect (MCQ only)
 - 2-mark questions: +2 for correct, -0.66 for incorrect (MCQ only)
 - 3-mark questions: +3 for correct, -0.66 for incorrect (MCQ only)
 - NAT questions carry no negative marking.
 - MSQ questions carry no negative marking.
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4. How to Answer

- For MCQ: Select exactly one option.
 - For MSQ: Select one or more options. Partial credit is NOT awarded.
 - For NAT: Enter the numerical answer in the provided space. Round off as specified.
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5. Rough Work

All rough work should be done only in the space provided in this question paper. Do not use additional sheets.

6. Calculator

An on-screen virtual calculator is available for use during the examination. Physical calculators are NOT permitted.

7. Important Notes

- Do not open the seal of the Question Booklet until you are instructed to do so.
 - The Question Booklet contains multiple pages. Verify the page count.
 - Immediately report any discrepancy in the Question Booklet to the invigilator.
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Q1. 1M [-0.33]

[General Aptitude]

10% of 500:

- (A) 5
- (B) 50
- (C) 500
- (D) 0.5

Q2. 1M [-0.33]

[Engineering Mathematics]

Rank of 3x3 zero matrix:

- (A) 0
- (B) 1
- (C) 2
- (D) 3

Q3. 1M [-0.33]

[Physical Metallurgy]

BCC coordination number:

- (A) 4
- (B) 6
- (C) 8
- (D) 12

Q4. 1M [-0.33]

[Mechanical Metallurgy]

Edge dislocation moves by:

- (A) Climb
- (B) Glide
- (C) Cross slip
- (D) None

Q5. 1M [no -ve]

[Extractive Metallurgy]

Reducing agent in blast furnace:

Enter your answer here

Q6. 1M [-0.33]

[Phase Transformations]

Eutectic point has:

- (A) One phase
 - (B) Two phases
 - (C) Three phases simultaneously
 - (D) Four phases
-

Q7.

1M [-0.33]

[Heat Treatment]

Martensite is formed by:

- (A) Slow cooling
- (B) Quenching
- (C) Annealing
- (D) Normalizing

Q8.

1M [no -ve]

[Corrosion]

Cathodic reaction in rusting:

Enter your answer here

Q9.

1M [no -ve]

[Welding & Joining]

Shielded metal arc uses:

Enter your answer here

Q10.

1M [no -ve]

[Non-Ferrous Metals & Alloys]

Duralumin is Al +:

Enter your answer here

Q11.

1M [-0.33]

[Non-Ferrous Metals & Alloys]

Duralumin is Al +:

- (A) Cu
- (B) Mg
- (C) Si
- (D) Mn

Q12.

1M [no -ve]

[Physical Metallurgy]

BCC coordination number:

Enter your answer here

Q13.

1M [-0.33]

[General Aptitude]

10% of 500:

- (A) 5
- (B) 50
- (C) 500
- (D) 0.5

Q14.

1M [-0.33]

[General Aptitude]

10% of 500:

- (A) 5
- (B) 50
- (C) 500
- (D) 0.5

Q15.

1M [-0.33]

[Phase Transformations]

Eutectic point has:

- (A) One phase
- (B) Two phases
- (C) Three phases simultaneously
- (D) Four phases

Q16.

1M [-0.33]

[Mechanical Metallurgy]

Edge dislocation moves by:

- (A) Climb
- (B) Glide
- (C) Cross slip
- (D) None

Q17.

1M [no -ve]

[Heat Treatment]

Martensite is formed by:

*Enter your answer here***Q18.**

1M [no -ve]

[Physical Metallurgy]

BCC coordination number:

*Enter your answer here***Q19.**

1M [-0.33]

[Physical Metallurgy]

BCC coordination number:

- (A) 4
- (B) 6
- (C) 8
- (D) 12

Q20.

1M [-0.33]

[General Aptitude]

10% of 500:

- (A) 5
- (B) 50
- (C) 500
- (D) 0.5

Q21.

1M [no -ve]

[Mechanical Metallurgy]

Edge dislocation moves by:

*Enter your answer here***Q22.**

1M [no -ve]

[Heat Treatment]

Martensite is formed by:

*Enter your answer here***Q23.**

1M [no -ve]

[Mechanical Metallurgy]

Edge dislocation moves by:

*Enter your answer here***Q24.**

1M [-0.33]

[Corrosion]

Cathodic reaction in rusting:

- (A) $\text{Fe} \rightarrow \text{Fe}^{2+} + 2\text{e}^-$
- (B) $\text{O}_2 + 2\text{H}_2\text{O} + 4\text{e}^- \rightarrow 4\text{OH}^-$
- (C) Both
- (D) None

Q25.

1M [no -ve]

[Extractive Metallurgy]

Reducing agent in blast furnace:

*Enter your answer here***Q26.**

1M [-0.33]

[Engineering Mathematics]

Rank of 3x3 zero matrix:

- (A) 0
- (B) 1
- (C) 2
- (D) 3

Q27.

1M [-0.33]

[Mechanical Metallurgy]

Edge dislocation moves by:

- (A) Climb
- (B) Glide
- (C) Cross slip
- (D) None

Q28.

1M [-0.33]

[Mechanical Metallurgy]

Edge dislocation moves by:

- (A) Climb
- (B) Glide
- (C) Cross slip
- (D) None

Q29.

1M [-0.33]

[Phase Transformations]

Eutectic point has:

- (A) One phase
- (B) Two phases
- (C) Three phases simultaneously
- (D) Four phases

Q30.

1M [-0.33]

[Extractive Metallurgy]

Reducing agent in blast furnace:

- (A) Coke
- (B) Limestone
- (C) Hot air
- (D) Slag

Q31.

2M [no -ve]

[Mechanical Metallurgy]

Edge dislocation moves by:

*Enter your answer here***Q32.**

2M [-0.66]

[Physical Metallurgy]

BCC coordination number:

- (A) 4
- (B) 6
- (C) 8
- (D) 12

Q33.

2M [-0.66]

[Non-Ferrous Metals & Alloys]

Duralumin is Al +:

- (A) Cu
- (B) Mg
- (C) Si
- (D) Mn

Q34.

2M [no -ve]

[Welding & Joining]

Shielded metal arc uses:

*Enter your answer here***Q35.**

2M [-0.66]

[General Aptitude]

10% of 500:

- (A) 5
- (B) 50
- (C) 500
- (D) 0.5

Q36.

2M [-0.66]

[Welding & Joining]

Shielded metal arc uses:

- (A) Gas shield
- (B) Flux coated electrode
- (C) Both
- (D) None

Q37.

2M [-0.66]

[Engineering Mathematics]

Rank of 3x3 zero matrix:

- (A) 0
- (B) 1
- (C) 2
- (D) 3

Q38.

2M [-0.66]

[Corrosion]

Cathodic reaction in rusting:

- (A) $\text{Fe} \rightarrow \text{Fe}^{2+} + 2\text{e}^-$
- (B) $\text{O}_2 + 2\text{H}_2\text{O} + 4\text{e}^- \rightarrow 4\text{OH}^-$
- (C) Both
- (D) None

Q39.

2M [-0.66]

[Engineering Mathematics]

Rank of 3x3 zero matrix:

- (A) 0
- (B) 1
- (C) 2
- (D) 3

Q40.

2M [no -ve]

[Heat Treatment]

Martensite is formed by:

*Enter your answer here***Q41.**

2M [-0.66]

[Engineering Mathematics]

Rank of 3x3 zero matrix:

- (A) 0
- (B) 1
- (C) 2
- (D) 3

Q42.

2M [-0.66]

[Non-Ferrous Metals & Alloys]

Duralumin is Al +:

- (A) Cu
- (B) Mg
- (C) Si
- (D) Mn

Q43.

2M [no -ve]

[Extractive Metallurgy]

Reducing agent in blast furnace:

- (A) Coke
- (B) Limestone
- (C) Hot air
- (D) Slag

Q44.

2M [-0.66]

[Phase Transformations]

Eutectic point has:

- (A) One phase
- (B) Two phases
- (C) Three phases simultaneously
- (D) Four phases

Q45.

2M [-0.66]

[Non-Ferrous Metals & Alloys]

Duralumin is Al +:

- (A) Cu
- (B) Mg
- (C) Si
- (D) Mn

Q46.

2M [no -ve]

[Welding & Joining]

Shielded metal arc uses:

*Enter your answer here***Q47.**

2M [-0.66]

[Phase Transformations]

Eutectic point has:

- (A) One phase
- (B) Two phases
- (C) Three phases simultaneously
- (D) Four phases

Q48.

2M [-0.66]

[Engineering Mathematics]

Rank of 3x3 zero matrix:

- (A) 0
- (B) 1
- (C) 2
- (D) 3

Q49.

2M [-0.66]

[Non-Ferrous Metals & Alloys]

Duralumin is Al +:

- (A) Cu
- (B) Mg
- (C) Si
- (D) Mn

Q50.

2M [-0.66]

[Heat Treatment]

Martensite is formed by:

- (A) Slow cooling
- (B) Quenching
- (C) Annealing
- (D) Normalizing

Q51.

2M [-0.66]

[Mechanical Metallurgy]

Edge dislocation moves by:

- (A) Climb
- (B) Glide
- (C) Cross slip
- (D) None

Q52.

2M [-0.66]

[Mechanical Metallurgy]

Edge dislocation moves by:

- (A) Climb
- (B) Glide
- (C) Cross slip
- (D) None

Q53.

2M [no -ve]

[Welding & Joining]

Shielded metal arc uses:

*Enter your answer here***Q54.**

2M [no -ve]

[Corrosion]

Cathodic reaction in rusting:

*Enter your answer here***Q55.**

2M [-0.66]

[General Aptitude]

10% of 500:

- (A) 5
- (B) 50
- (C) 500
- (D) 0.5

Q56.

2M [no -ve]

[Corrosion]

Cathodic reaction in rusting:

*Enter your answer here***Q57.**

2M [no -ve]

[Heat Treatment]

Martensite is formed by:

*Enter your answer here***Q58.**

2M [-0.66]

[Physical Metallurgy]

BCC coordination number:

- (A) 4
- (B) 6
- (C) 8
- (D) 12

Q59.

2M [-0.66]

[Mechanical Metallurgy]

Edge dislocation moves by:

- (A) Climb
- (B) Glide
- (C) Cross slip
- (D) None

Q60.

2M [no -ve]

[Heat Treatment]

Martensite is formed by:

*Enter your answer here***Q61.**

2M [-0.66]

[Physical Metallurgy]

BCC coordination number:

- (A) 4
- (B) 6
- (C) 8
- (D) 12

Q62.

2M [no -ve]

[Corrosion]

Cathodic reaction in rusting:

*Enter your answer here***Q63.**

2M [no -ve]

[Corrosion]

Cathodic reaction in rusting:

*Enter your answer here***Q64.**

2M [no -ve]

[Mechanical Metallurgy]

Edge dislocation moves by:

*Enter your answer here***Q65.**

2M [no -ve]

[Extractive Metallurgy]

Reducing agent in blast furnace:

Enter your answer here